

Viscoelastic response of a tidally loaded hydrostatically prestressed spherical shell.

Reference configuration

The perfect undeformed sphere constitutes the reference configuration κ_0 described by radially dependent reference density ρ_0 , gravity potential V_0 , and hydrostatic prestress $\boldsymbol{\tau}_0$ (i.e., $\boldsymbol{\tau}_0 = -p_0 \mathbf{I}$, where p_0 denotes the mechanical hydrostatic pressure), we write the following

$$\begin{aligned} -\nabla p_0 + \rho_0 \vec{g}_0 &= 0, \\ \nabla^2 V_0 - 4\pi G \rho_0 &= 0, \end{aligned}$$

where G denotes the gravitational constant and $\vec{g}_0 = -\rho_0 \nabla V_0$ (and $\vec{g}_0 = -g_0 \vec{e}_r$). We require the continuity of the initial traction, gravitational potential, and gravitational acceleration at the surface, the tangential stress shall vanish at the CMB boundary (free-slip).

Non-compressible model

Let us move to the present configuration κ_t (small deformation of the sphere), the system is described by the following set of PDEs written within the framework of the Eulerian description

$$\nabla \cdot \boldsymbol{\tau}^E - (\nabla \rho_0 \cdot \vec{u}) \vec{g}_0 - \rho_0 (\nabla V_g + \nabla V_{\text{tide}}) = 0, \quad (1)$$

$$\boldsymbol{\tau}^d - \mu (\nabla \vec{u} + \nabla^T \vec{u}) = -\frac{\mu}{\eta} \int_0^t \boldsymbol{\tau}^d dt', \quad (2)$$

$$\nabla \cdot \vec{u} = 0, \quad (3)$$

$$\boldsymbol{\tau}^E \cdot \vec{e}_r = u_r \Delta \rho^{(\text{top})} \vec{g}_0, \quad (4)$$

$$-\boldsymbol{\tau}^E \cdot \vec{e}_r = u_r \Delta \rho^{(\text{bot})} \vec{g}_0 - \rho_{\text{core}} (V_g + V_{\text{tide}}), \quad (5)$$

where $\boldsymbol{\tau}^E$ denotes the incremental Eulerian Cauchy stress tensor, $\boldsymbol{\tau}^d$ is its deviatoric part, μ shear modulus, η viscosity, $\Delta \rho$ density jumps on the CMB boundary and surface, respectively, $\vec{u}$ is the displacement, ρ_{core} denotes the density of the liquid core, u_r radial displacement, and $\vec{e}_r$ is the radial unit vector. The driving tidal potential is denoted as V_{tide} and reads

$$V_{\text{tide}}(\vec{r}, t) = \frac{3}{2} \sqrt{\frac{2\pi}{15}} \Omega^2 r^2 \theta_0 [\exp(i\Omega t) Y_{21}(\theta, \phi) - \exp(-i\Omega t) Y_{2-1}],$$

and the incremental gravity potential V_g satisfies

$$\nabla^2 V_g - 4\pi G \rho^E = 0,$$

where $\rho^E = -\nabla \cdot (\rho_0 \vec{u})$ is the Eulerian increment of density. The additional term representing density jumps inside the mantle is decomposed as follows

$$\begin{aligned} \vec{Y}_{jm}^{j-1} : & \quad -g_0 |\nabla \rho_0| \left(\frac{j}{2j+1} u_{jm}^{j-1} - \frac{\sqrt{j(j+1)}}{2j+1} u_{jm}^{j+1} \right), \\ \vec{Y}_{jm}^{j+1} : & \quad -g_0 |\nabla \rho_0| \left(-\frac{\sqrt{j(j+1)}}{2j+1} u_{jm}^{j-1} + \frac{j+1}{2j+1} u_{jm}^{j+1} \right). \end{aligned}$$

Compressible model

We remain in the present configuration κ_t , however, we turn on the compressibility of the viscoelastic shell. As a result, the governing equation $\nabla \cdot \vec{u}$ no longer holds and an additional term pops up in the equation of motion

$$\nabla \cdot \boldsymbol{\tau}^E - (\nabla \rho_0 \cdot \vec{u}) \vec{g}_0 - (\nabla \cdot \vec{u}) \rho_0 \vec{g}_0 - \rho_0 (\nabla V_g + \nabla V_{\text{tide}}) = 0$$

Thus, the reference density ρ_0 now changes also due to the non-zero divergence of the displacement field. The incremental Eulerian and Lagrangian Cauchy stress tensors are related (for hydrostatically prestressed continuum) by $\boldsymbol{\tau}^L = \boldsymbol{\tau}^E + u_r g_0 \rho_0 \mathbf{I}$, the Eulerian viscoelastic constitutive law reads

$$(\boldsymbol{\tau}^E + u_r g_0 \rho_0 \mathbf{I}) - \lambda (\nabla \cdot \vec{u}) - \mu (\nabla \vec{u} + \nabla^T \vec{u}) = -\frac{\mu}{\eta} \int_0^t (\boldsymbol{\tau}^E + u_r g_0 \rho_0 \mathbf{I}) dt'.$$

The trace of this equation yields

$$-\frac{1}{3}\text{Tr } \boldsymbol{\tau}^E + u_r g_0 \rho_0 + \frac{3\lambda + 2\mu}{3}(\nabla \cdot \vec{u}) = \frac{1}{\eta_B} \int_0^t (\boldsymbol{\tau}^E + u_r g_0 \rho_0 \mathbf{I}) dt'.$$

The Eulerian incremental Cauchy stress tensor can be decomposed into pressure and its deviatoric part ($\boldsymbol{\tau}^E = -p^E \mathbf{I} + \mathbf{D}$), spectral decomposition then gives

$$\begin{aligned} \boldsymbol{\tau}^E &= \sum_{j=0}^{+\infty} \sum_{m=-j}^j \tau_{jm}^{j0}(r) \mathbf{Y}_{jm}^{j,0}(\theta, \phi) + \sum_{j=0}^{+\infty} \sum_{m=-j}^j \sum_{k=|j-2|}^{j+2} \tau_{jm}^{k,2}(r) \mathbf{Y}_{jm}^{k,2}(\theta, \phi), \\ p^E &= \sum_{jm} p_{jm}^E(r) Y_{jm}(\theta, \phi), \end{aligned}$$

however, one should keep in mind that coefficients p_{jm}^E and τ_{jm}^{j0} are not the same. By projecting $-p^E \mathbf{I}$ into the direction of $\vec{e}_r$, we will be able to find the relation between coefficients τ_{jm}^{j0} and p_{jm}^E , thus

$$\begin{aligned} (-p^E \mathbf{I}) \cdot \vec{e}_r &= \sum_{j=0}^{+\infty} \sum_{m=-j}^j \tau_{jm}^{j0} (\mathbf{Y}_{jm}^{j,0} \cdot \vec{e}_r) = \sum_{j=0}^{+\infty} \sum_{m=-j}^j \tau_{jm}^{j0} \left(\sqrt{\frac{j+1}{3(2j+1)}} \vec{Y}_{jm}^{j+1} - \sqrt{\frac{j}{3(2j+1)}} \vec{Y}_{jm}^{j-1} \right), \\ (-p^E) \vec{e}_r &= \sum_{j=0}^{+\infty} \sum_{m=-j}^j (-p_{jm}^E) Y_{jm}(\theta, \phi) \vec{e}_r = \sum_{j=0}^{+\infty} \sum_{m=-j}^j (-p_{jm}^E) \left(\sqrt{\frac{j}{2j+1}} \vec{Y}_{jm}^{j-1} - \sqrt{\frac{j+1}{2j+1}} \vec{Y}_{jm}^{j+1} \right). \end{aligned}$$

We can see that the coefficients are related by

$$p_{jm}^E = -\frac{\tau_{jm}^{j0}}{\sqrt{3}}.$$

The continuity equation (under the assumption $\eta_B \gg \eta$) transforms into

$$\begin{aligned} \frac{\tau_{jm}^{j0}}{\sqrt{3}} + \sqrt{\frac{j}{2j+1}} \frac{3\lambda + 2\mu}{3} \left(\frac{d}{dr} - \frac{j-1}{r} \right) u_{jm}^{j-1} - \sqrt{\frac{j+1}{2j+1}} \frac{3\lambda + 2\mu}{3} \left(\frac{d}{dr} + \frac{j+2}{r} \right) u_{jm}^{j+1} \\ - \sqrt{\frac{j}{2j+1}} \rho_0 g_0 u_{jm}^{j-1} + \sqrt{\frac{j+1}{2j+1}} \rho_0 g_0 u_{jm}^{j+1} \approx 0. \end{aligned}$$

And the spectral decomposition of the term $-(\nabla \cdot \vec{u}) \rho_0 \vec{g}_0$ (under the assumption $\eta_B \gg \eta$) reads

$$\begin{aligned} \vec{Y}_{jm}^{j-1} : \quad & \frac{\rho_0^2 g_0^2}{\lambda + \frac{2\mu}{3}} \left(\frac{j}{2j+1} u_{jm}^{j-1} - \frac{\sqrt{j(j+1)}}{2j+1} u_{jm}^{j+1} \right) - \frac{\tau_{jm}^{j0}}{\sqrt{3}} \sqrt{\frac{j}{2j+1}} \frac{\rho_0 g_0}{\lambda + \frac{2\mu}{3}} \approx 0, \\ \vec{Y}_{jm}^{j+1} : \quad & \frac{\rho_0^2 g_0^2}{\lambda + \frac{2\mu}{3}} \left(-\frac{\sqrt{j(j+1)}}{2j+1} u_{jm}^{j-1} + \frac{j+1}{2j+1} u_{jm}^{j+1} \right) + \frac{\tau_{jm}^{j0}}{\sqrt{3}} \sqrt{\frac{j+1}{2j+1}} \frac{\rho_0 g_0}{\lambda + \frac{2\mu}{3}} \approx 0. \end{aligned}$$

So, the modified PDEs can be written down (in reality, we set $\eta_B \rightarrow +\infty$) as follows

$$\nabla \cdot \boldsymbol{\tau}^E - (\nabla \rho_0 \cdot \vec{u}) \vec{g}_0 - (\nabla \cdot \vec{u}) \rho_0 \vec{g}_0 - \rho_0 (\nabla V_g + \nabla V_{\text{tide}}) \approx 0, \quad (6)$$

$$\boldsymbol{\tau}^d - \mu (\nabla \vec{u} + \nabla^T \vec{u}) = -\frac{\mu}{\eta} \int_0^t \boldsymbol{\tau}^d dt', \quad (7)$$

$$-\frac{1}{3}\text{Tr } \boldsymbol{\tau}^E + u_r g_0 \rho_0 + \frac{3\lambda + 2\mu}{3}(\nabla \cdot \vec{u}) \approx 0, \quad (8)$$

$$\boldsymbol{\tau}^E \cdot \vec{e}_r = u_r \Delta \rho^{(\text{top})} \vec{g}_0, \quad (9)$$

$$-\boldsymbol{\tau}^E \cdot \vec{e}_r = u_r \Delta \rho^{(\text{bot})} \vec{g}_0 - \rho_{\text{core}} (V_g + V_{\text{tide}}). \quad (10)$$

Love Numbers

The (tidal) Love numbers describing the viscoelastic response

$$k_2^T = \frac{V_{\text{surf}}}{V_{\text{tide}}} \quad h_2^T = \frac{u_r^{\text{top}} g_0}{V_{\text{tide}}} \quad l_2^T = \frac{u_t^{\text{top}} g_0}{6V_{\text{tide}}}.$$